

1.5B people keep 9T+ photos in Google Photos, and retrieval breaks at interpretation and surfacing, almost never at recovery

1.5B+
monthly Google Photos users

9T+
photos and videos stored

Goal (brief)
"Increase the percentage of users who successfully retrieve a photo they remember but cannot precisely describe."

North Star URR: users with a vague-memory search in 28 days who reach the photo

URR = E × [1 − (1 − I × S × R)^ñ]

Term	Brief's question	Engine (144)	Survey (15)	Reported
E Expression	Can't express it?	2 (1.4%)	3 (20%)	2 of 3 never type
I Interpretation ★	Clues misread?	51 (35.4%)	3 (20%)	"Specify the year"
S Surfacing ★	Photo in results at all?	60 (41.7%)	5 (33%)	Target absent
R Recognition	Results hard to judge?	2 (1.4%)	2 (13%)	7 of 15 unsure
ñ Recovery	Can't refine?	1 (0.7%)	1 (7%)	2–3 tries

Baseline URR: 40% (self-reported proxy, no telemetry). 6 of 15 survey respondents were sure they found the photo; 0 of 5 MVP testers found theirs in their own Photos.

What the decomposition says

- 77% in two terms: I + S = 111 of 144 failures, so the product acts there.
- Recovery is out: 0.7% of failures, 1 of 15 in the survey; kept as a safety net and diagnostic.
- Where Google invested: Ask Photos reads prompts better but doesn't group moments or resolve "last Diwali".

Browse path (outside the formula)

10 of 15 scrolled or opened an album first; MVP testers last found old photos by scroll 3, album 2, search 0.

The engine turns 85,140 public posts into 144 retrieval attempts, each recording what was remembered, what was forgotten, and where search broke

1 · COLLECT

85,140

Public posts
(Play Store,
App Store, YT,
Reddit)

2 · SCREEN

819

Relevant (Gate
A 5,305 → Gate
B 1,333
sample)

3 · EXTRACT

144

Specific
attempts (720
episodes; 62
scoreable)

4 · AUDIT

203

Pairs checked
blind by Qwen
27B model

5 · COMPARE

9 Areas

Ranked only on
agreed fields
(O1–O9)

Extracted Field in Schema	The Brief's Underlying Question Answered	Reliability Check
asset_type	What kinds of old photos do users struggle to retrieve?	Validated in Gate B
cues_retained (14 cue types)	What do people actually remember about the photo?	Jaccard 0.557
cues_lost (date, place, album, words)	What have they forgotten when search fails?	Jaccard 0.673
query_verbatim · search_mode	How do they search when memory is incomplete?	Quote verify 85.2%
failure_stage (5 stages)	Where does retrieval break? (Part 3 decomposition)	κ 0.509 (Moderate)

Why This Is More Than Summarisation

1. Structure, not sentiment: Every review was converted into a structured, typed record with fixed vocabularies, not qualitative impression tags.

2. Pre-registered rules: Hypotheses H1–H5 and decision thresholds were set before extraction, so the data could not be read to fit them.

3. The audit overruled the ranking: When the blind audit showed hypothesis agreement was low (Jaccard 0.347) while failure stage agreement was solid (κ 0.509), the engine demoted hypothesis ranking and pivoted to failure stages.

retrieval-discovery-engine.vercel.app

People remember roughly when, but search needs the exact date, and the exact date is what they forget

The Brief's Four Questions Answered (Engine vs. Survey)

Question	Engine (144 Specific Attempts)	Survey (n=15 / 14 Dedup)
Photo Kinds	Personal photos 80 · multi-photo 26 · videos 16 · screenshots 7 · docs 3	All 3 real-trouble cases were document or medicine photos
Remembered	Roughly when 40 · object 17 · exact date 14 · text 12 · people 10	Roughly when 8 · object 7 · people with me 6
Forgotten	The date 37 · album 29 · exact words 10 · place 10	When it was taken 9 · words to search 9
Search Method	20 of 32 queries are 1 word ("dog"); 10 name a time ("Halloween 2024")	First move: scroll 9/15; own-app: 4 of 5 strip time to bare nouns

Opportunity Areas: Why O1 Leads

O1 (Rough time or an event): 42 → 37 after exclusions (18 never surfaced, 12 misread).

O8 (No clue at all): 47 attempts, larger, but there is no memory for search to use. O1 is the largest group where memory exists and search ignores it.

Excluded: O7 exact date known (14) · O9 path moved by an update (12).

Every Pre-Registered Hypothesis Gets a Verdict

H1 Episodic time: SUPPORTED (root cause).
Roughly when is the #1 kept cue (40); the date is the #1 lost cue (37).

H2 Recognition: SUPPORTED (secondary). 41.7% never surfaced; 7 of 15 ended unsure.

H3 Dead-end recovery: REFINED, NOT THE LEAD. Overruled by the audit: recovery is 0.7% of failures (stage κ 0.509).

H4 Hinglish: NOT TESTED. Weak signal (2 of 15 survey, 1 of 6 tests); kept as a design choice.

H5 Text not indexed: WEAKLY SUPPORTED (3 docs; the index scores 0.000 on text in images).

H6 Path changed: PRESENT BUT MINOR (8.3%), added post hoc.

Most people scroll before they search, and nearly half end close but unsure

Research Methodology Rationale

Structured self-serve questionnaire (n=15, 23–30 Sep): every answer maps onto the engine's vocabulary.

Why: 100% scoreable stories (vs 8.6% of public posts); reached 15 people vs 2 call contacts.

Costs: self-reported, convenience sample; 8 of 15 fit the segment.

[Survey questions ↗](#)

Survey vs. Engine (Side-by-Side)

Metric	Engine (144)	Survey (15 / 14)
Most remembered	Roughly when (40)	Roughly when (8)
Ended unsure	Not measurable	7 of 15 (47%)
First move: scroll	15 workarounds	9 of 15 (60%)

Why it matters to them

Trips and outings, a photo for LinkedIn, a medical report.
12 of 15 spent 5+ minutes, past the brief's five-minute bar; 3 had to ask someone or get the document again.

Reported Retrieval Incidents (Survey Respondents)

ID	Target Photo	What They Remembered	What They Typed	Break	Outcome
S04	Wedding photo	Roughly when, event, people	"wedding, pichle saal diwali"	Interpretation	Unsure
S03	Personal photo	Roughly when, event, object	"gym"	Recognition	Unsure
S08	Personal photo	Roughly when, event	(Didn't know what to type; scrolled)	Expression	Found
S12	Personal photo	Roughly when, event	(Never typed; opened album)	Expression	Unsure
S06	Medicine bill	Object, text in bill	"medicine, bill"	Surfacing	Never found

"wedding, pichle saal diwali" — S04 (Hinglish: a festival and relative year, not a calendar date)

"medicine, bill" — S06 (Never found; had to get the document again)

Ask Photos: 7 of 15 had never heard of it; all 4 who used it got "related photos, but not the one".

The segment is people who can place the moment only roughly, and search can't turn "roughly" into a date window

Chosen Target Segment

Long-time Google Photos users searching for a personal moment they can place only roughly.

Defined by **cognitive memory state** (what they remember), not demographics: "around Diwali", "my sister's graduation", "last winter".

Segment Sizing Filter	Engine (144)	Survey (15)
Kept rough time or event (O1)	42 (29%)	8 (53%)
Target Segment (After Exclusions)	37 (26%)	8 (53%)
Personal photo or video	30 of 37	All 8
Large library (5,000+ items)	Not measured	5 of 8

42 → 37: minus 2 who also knew the exact date (plain search works) and 3 whose browse path moved in an update (H6). Sync and corruption issues are out of scope.

The Root Cause: The Broken Link

Human Memory Stores
EPISODES

Rough time, event anchors, surrounding scenes ("last winter", "cousin's wedding")



Google Photos Indexes
ITEMS & DATES

Isolated photos with calendar timestamps (YYYY-MM-DD) and tags

"I'll type in something super simple, like 'Halloween 2024' and it seriously can't find anything?" — Play Store review

Asking why, for the 37 segment attempts:

- 1. **Why isn't the photo found?** It never surfaces (18 of 37).
- 2. **Why not?** Search misreads the clue (12): "last Diwali" is neither a keyword nor a date.
- 3. **Why does that matter?** People keep the event and lose the date (the #1 lost cue, 37 of 144).
- 4. **Root cause:** nothing turns "roughly when" into a date window.
Real app: 4 of 5 testers typed bare nouns ("vacation", "gym"); 0 of 5 found the photo.

People already scroll to "roughly when" by hand and still end unsure, so the product should take that step for them

PROBLEM DEFINITION

"When people want a photo they can place only roughly ("last Diwali"), they can describe the moment but can't make Photos show it: search needs a date or keyword they've forgotten. So they scroll by hand, and nearly half end unsure."

The Workaround is the Design Blueprint

Engine: 15 of 21 workarounds are manual scrolling (7 of 10 failed): *"Every time I scrolled to the right time frame, it kept loading more pictures..."*

Survey: Scrolling was the first move for 9 of 15. 5 of those 9 still ended unsure.

Insight: people already retrieve by moment, by hand.
The product should find the time window and group the moments.

Solution Rationale: Why This, and Why Google Wins

Option	How you search
Apple Photos (iOS 18)	Describe what's in the photo
Ask Photos	Ask Gemini; all 7 who tried got "related, not the one"
Memory Trails	Roughly when → date window → moments, with why

Why Google can win: it already holds every user's dates, places and faces, and could read events from Calendar. No standalone app has that data.

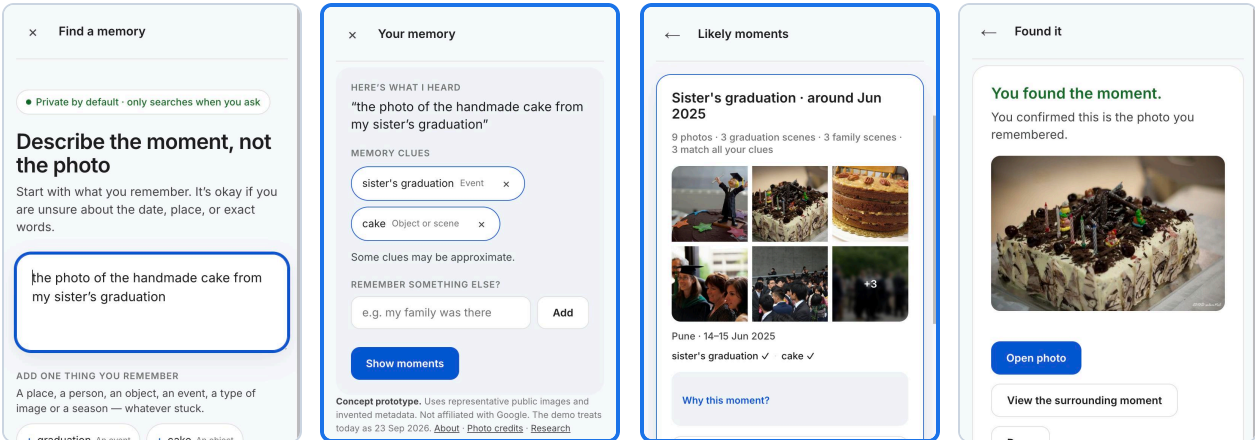
HOW THE THINKING EVOLVED

1. **Metric:** per-search success → **per-user URR**
2. **Outcomes:** recovery agent → **5 stages + browse**
3. **Discovery:** H3 ranked top → **overturned by audit**
4. **Reported behaviour:** retry search → **scroll by hand**
5. **Problem:** "search is bad" → **items vs. moments**

Memory Trails turns "roughly when and where" into moments you can recognise

Placement & flow

Where it lives: inside Photos search. When a search fails, *"Can't describe it?"* opens it with the words carried over.



1 · Describe 2 · Clues 3 · Moments 4 · Confirm

memory-trails-v2.vercel.app ↗

Blue frame = AI step

What's new vs Photos search

- 1. Clues → date window: "pichle saal Diwali"
- 2. Moments, not a grid: "3 match all your clues"
- 3. "Why this moment?": matched vs approximate
- 4. Soft scoring: a wrong date lowers rank, never hides

Empirical Evaluation Ladder (120 Tasks, 1,282 Photos)

Cue Dropout Level	What the Query Keeps	Plain CLIP	Memory Trails	Moment@5
L3 (Fully described)	Vague time + exact place + library word	0.172	0.962	0.933
L2 (Two vague cues)	Paraphrased content + 2 vague cues	0.209	0.276	0.333
L1 (One vague cue)	Paraphrased content + 1 vague time cue	0.242	0.309	0.333
L0 (Content only)	Pure scene description, no metadata	0.312	0.312	0.267
Weighted by real memory	Each level weighted by how often the 144 real attempts keep it	0.259	0.334 (+0.075)	0.336 (+0.111)

Read this first: L3's 0.962 needs three cues; only 4% (6 of 144) of real attempts kept that many. Weighted by real memory, recall gains **+0.075** and moment@5 **+0.111**: grouping into moments helps more than flat ranking.

Scope: tests the re-entry interaction on a 1,282-photo Creative Commons library, not production-scale accuracy.

5 of 6 testers were sure they found the photo and 4 of 5 found it easier than Google Photos, but compound dates still trip it up

ID	Own Photos search → result	Task 1 (cat)	Task 2 (dog)	Sure	vs GP	Would use
R01	"Wedding before COVID" → wrong years	Found, sure	Not found	5/5	5/5	Every time
R02	"gym" → can't tell which	Found, sure	Found (1-tap alt)	4/5	4/5	When fails
R03	(Apple Photos user)	Found, sure	Found (nearby)	5/5	—	Every time
R04	"vacation" → wrong years	Found, sure	Found (1-tap alt)	4/5	4/5	When fails
R05	"Wedding event" → can't tell which	Found, sure	Found (clue edit)	4/5	4/5	Every time
R06	"Kerala photos" → can't tell which	Right day, unsure	Found (1-tap alt)	4/5	3/5	Every time

6 of 6

Fit the segment

4.33 / 5

Task 1 sureness

4.0 / 5

Ease vs Photos

0 of 5

Found it in own Photos

MVP test questions ↗

memory-trails-v2.vercel.app ↗

What testers said

"Shows the moment, not a grid" — R02

"Moving to nearby months, like scrolling but faster" — R03

"Works without knowing what to type" — R04

"Finds exact month & year... understands mix of Hindi and English" — R06

What broke → next iteration

1. Compound time: R01 stuck on "pichle ke pichle saal" (year before last). Fix: parse compound offsets; editable year.

2. Clues missed: R04 didn't notice them. Fix: high-contrast "Showing Diwali 2025 · Tap to change".

3. Speed: R05, R06 wanted faster loading. Fix: pre-cache moments.

Success is URR rising from 40% to 45% in a user-level A/B test, read early through one leading metric per URR term

North Star: 28-day URR

Definition: share of users with a vague-memory search in 28 days who reach the photo. Counts users, not searches.

Baseline 40% → target 45%. The baseline is a self-reported proxy (6 of 15 sure they found it). +5 pts is the smallest lift worth shipping.

Test: 8 weeks, randomised by user; ~1,600 users per arm for 80% power at α 0.05.

Already measured on the MVP

Recall@20 weighted by real memory **0.259 → 0.334** · moment@5 **+0.111** · tester sureness **4.33 / 5** · ease vs Photos **4.0 / 5**

Decision rule

Ship if URR rises ≥ 5 pts and no guardrail trips. **Extend** if the lift is positive but under 5. **Stop** if false confirmation passes 3%.

Leading metrics: one per URR term

Metric	Term	Target
Entry rate after a failed search	Expression	$\geq 12\%$
Clue correction (chip edited)	Interpretation	15–25%
"Not in any of these?" rate	Surfacing	Falling
Confirm after opening a moment	Recognition	$\geq 55\%$

Guardrails: stop or roll back

False confirmation (confirmed, then kept searching)	$< 3\%$
P95 time to first moment	< 1.5 s
Abandonment before the first moment	No rise vs control

Diagnostics: why it moved

Recall@20 by cue type · near-miss recovery rate · "Why this moment?" views before confirming

The biggest risk is that the win sits where memory is richest, and the evidence is small and self-reported

Risks & mitigations

Risk	Mitigation & evidence
R1. The win sits where memory is richest (+0.79 at L3, +0.07 at L1)	Lean on moment recognition (moment@5 0.333 at L1–L2); report L1 separately in the A/B.
R2. Date filters hide photos	Soft scoring built: L3 recall 0.851 → 0.962.
R3. False confirmation (worst failure)	Never auto-confirm; show the surrounding photos; stop launch above 3%.
R4. Can't read text in photos (0.000 on receipts)	Add OCR text to the index.
R5. Ask Photos may already suffice	4 of 4 survey and 3 of 3 test users got "related photos, not the one".
R6. Sensitive memories (medical, relationships)	User-initiated only; the session persists nothing.

Limitations, stated plainly

- Play Store is 86.2% of engine posts
- Survey n = 15 vs a target of 30, self-reported
- MVP tests n = 6, unmoderated, no session logs
- URR baselines modelled; no telemetry
- Eval library: 1,282 CC photos, synthetic episodes
- Ranking weights are judgement
- Hinglish (H4) is a weak signal only

Next iteration

1. Parse compound dates ("pichle ke pichle saal")
2. Make the active date clue unmissable
3. Run the A/B in one market